



University of Tartu

Practical work 7: Advanced SQL (Basketball Edition)

Exercise 1: View Creation

Run the script (from Moodle) to create the `Team`, `Player`, `Coach`, and `Match` tables in Postgres.

A. View creation

1. Create a view named `team_summary` allowing the sports director to verify team identities (team name, city, coach). Check view content and drop it.
 2. Create `high_scorers` view on the `Player` table grouping the players whose average points per game are higher than the overall average and whose position corresponds to 'Guard'. Check your results.
 3. Create `young_players` view on the `Player` table grouping male players whose age is less than 23. The insertion or update on this view must always respect this constraint.
 4. Set to 21 the age of the players in the `young_players` view whose first name is 'Luca'.
 5. Create a `team_roster` view containing the details of the players in the 'Lyon Eagles' team.
 6. Add a new male player aged less than 22 to the `Player` table. Check in the view table if the maintainability has been reflected.
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Exercise 2: Queries

B. Queries

1. What are the different player positions? The list must not contain duplicates.

```
SELECT DISTINCT position FROM Players
```

2. Which players belong to a team whose city name ends with 'ville'? Sort the result in descending order by age.

3. Who are the players coached by 'Jean Dupont'? Sort the result by team name.
 4. What is the minimum and maximum age of players?
 5. Which players are neither 'Forward', nor 'Center', nor 'Coach Assistant'? Write the query using subqueries.
 6. Who is the youngest player contained in the database? Display their name, age, position, and average points.
 7. What is the average number of points scored by players who play as 'Forward'?
 8. Calculate the number of players in the league and the total average points per game for each position.
 9. Modify the previous query to get the result also by team in each position.
 10. Calculate the number of players per team. Display also the team name and coach name.
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Exercise 3: More Queries

C. More queries

1. Show the average number of points for each team with at least 8 players and order the results by average points in descending order.
2. Find players whose average points are higher than the average points in their team.
3. List players who play in 'Lyon' or 'Marseille'.
4. Find players who are not assigned to any match.
5. Find players whose average points are above the overall league average.
6. Find players who have played in at least one match coached by 'Jean Dupont'.
7. Show 5 player names in uppercase and how long their email address is.
8. Rank players within each team based on their average points, with the highest scorer getting rank 1.
9. Find players whose email ends with '@nba.com'.